

A Two-Stage Decision Support Model for Cut Flower Palletisation in Full Charter Air Freight: Integer Programming for Allocation and Online Bin Packing for Placement

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Abstract

Full charter air freight is the dominant transport mode for exporting cut flowers from the Andean region to the United States. Because the aircraft is contracted at a fixed price, every unused cubic metre is an unrecoverable cost rather than a saved tariff. This study addresses the Air Cargo Palletisation Problem (APP) for an Airbus A330-200F operating the Bogotá–Miami corridor on behalf of a major South American flower exporter. Existing decompositions of the Air Cargo Load Planning Problem solve the palletisation subproblem offline, assuming that the complete manifest is available when each box is placed. That assumption does not hold at the origin terminal, where boxes reach the build-up conveyor sequentially and placement is irreversible. We propose a two-stage decision support model that partitions the problem by information horizon rather than by geometry. Stage 1 is an integer linear program that assigns boxes to the 32 pallet positions of the aircraft subject to volumetric capacity, pallet and aircraft weight limits, and a lateral balance bound, and whose extensions additionally shape the dimensional composition of each pallet’s manifest. Stage 2 embeds the resulting per-pallet manifests in the DeepPack3D framework and places boxes using deterministic online heuristics and a deep reinforcement learning agent under a finite lookahead window and a gravitational stability requirement. The stages are coupled through empirically calibrated

fill factors. On the historical charter dataset of 8,500 boxes, the exact allocation stage assigns 7,967 boxes (64,986 kg, 392.6 m³), of which the best online policy (Bottom-Left, lookahead $k = 7$) physically places $88.4\% \pm 0.7\%$ across random arrival sequences — demonstrating that volumetric feasibility at the allocation level does not guarantee geometric packability, and quantifying that gap for the first time at aircraft scale. Coupling the stages through empirically calibrated fill factors closes most of that gap in a single iteration, raising the physically realised load from 57.0 to 63.3 tonnes: +12.5% over the historical manual load of 56.3 tonnes, equivalent to an 11.1% reduction in unit freight cost (against the 13.3% that the allocation stage alone would overpromise). A single-stage online benchmark packs more densely but violates aircraft and per-position weight limits in every replication, establishing the admissibility value of the allocation stage. A sensitivity analysis over the degree of upstream sequence control shows its marginal value to be small (≤ 1.4 percentage points) relative to allocation quality, while penalising dimensional heterogeneity within pallets raises packability to 97.2% at essentially zero allocation cost.

Keywords: air cargo palletisation; cut flower supply chain; integer linear programming; online 3D bin packing; weight and balance; decision support model; Airbus A330-200F

1 Introduction

Cut flowers are among the few high value agricultural products in which low and middle income producers compete successfully in international markets, and the competitive position of Colombia and Ecuador rests almost entirely on air freight. The United States sources most of its imported fresh flowers from these two countries: in 2005 Colombia supplied 59% and Ecuador 18% of United States flower imports, and Miami International Airport, which receives about two thirds of those flowers, is the centre of their distribution (Vega, 2008a, pp. 16, 22). Bogotá–Miami is among the busiest extra-regional routes out of Latin America by offered capacity, although Ecuadorian exports split more evenly between Europe and the United States (Fioravanti et al., 2022, pp. 16, 36, 80). The product itself imposes the constraint: cut flowers lose quality and value continuously from harvest as a function of time, temperature and humidity (De Keizer et al., 2015, p. 61), and a consignment that falls below the quality level its buyer requires is rejected and written off rather than discounted (Rong et al., 2011, pp. 425–426). Speed is therefore not a service upgrade but a precondition of the business, and air transport, despite its cost, is the only mode that satisfies it (Vega, 2008a; Van Liemt, 1999).

That cost is substantial. Transport costs reach a quarter of the wholesale unit price of a stem in the United States and a third in Europe (Vega, 2008a, pp. 19–20), and logistics costs reach 40–60% of stem production cost in Kenyan floriculture (Button, 2020, p. 4), so

margins are exposed to fuel prices, capacity availability and seasonal peaks. Palletisation itself is priced and time consuming: it costs between 0.04 and 0.32 US dollars per box and takes about six hours out of the freshness window (Vega, 2008a, p. 14), which is the stage this study models. The industry response during peak season is the full charter agreement, in which the exporter, usually through a freight forwarder, contracts the entire aircraft at a fixed price. This changes the economics of loading in a way that is easy to state and difficult to exploit. Under a shared cargo agreement the exporter pays by chargeable weight, so unused space is the carrier’s problem. Under a full charter the cost is sunk at the moment of contracting, and the unit cost per box is simply the charter price divided by the number of boxes actually flown. Space that is not filled is money already spent and not recovered.

The operational problem is that filling it is genuinely hard. The sponsor company of this study is a vertically integrated grower with farms in Colombia, Ecuador and Kenya, operating seventeen post-harvest facilities that pack to customer specification. This customisation is a commercial asset and a logistical liability: the company handles dozens of distinct box configurations, and a single charter flight typically carries several thousand boxes drawn from twenty or more of them. At the export terminal, boxes arrive on a conveyor and are stacked by hand onto pallets by airline personnel working under a first come, first loaded discipline. The Head of Post-Harvest Operations of the sponsor company described the result plainly: it resembles a game of Tetris, and it is both slow and inefficient.

Beyond cost, the load factor has an environmental dimension. A charter is contracted as a whole: the aircraft, the route and the schedule are fixed before a single box is loaded, and what the exporter chooses is how much cargo leaves on that flight. Carrying more per flight reduces the number of charters needed for a given seasonal volume, and the emissions avoided are those of the flights not operated. Fuel burn does rise with the weight carried (Brueckner and Abreu, 2017, pp. 15–16), so the fall in carbon intensity per kilogram is smaller than the gain in payload, but it does not vanish. Improving utilisation is thus simultaneously a cost and an emissions lever.

Two prior studies commissioned by the same sponsor company approached this problem from opposite ends. The first formulated the allocation of boxes to the fixed pallet positions of an A330-200F as an integer linear program, maximising assigned weight and volume subject to pallet capacities and a lateral weight balancing requirement, and solved it exactly. The second treated the terminal reality directly, modelling the build-up conveyor as an online three-dimensional bin packing problem with limited lookahead and irreversible placement, and evaluating deterministic heuristics at aircraft scale. Each study answered half of the question. The first decided which boxes go to which pallet but said nothing about whether the assigned set could physically be stacked in the order it arrived. The second decided where each box goes inside a pallet but took the arrival

stream as exogenous and ignored weight and balance entirely. The present work integrates them.

The integration is not merely additive, and the reason is worth stating precisely. The two studies appear to be in methodological conflict, one assuming complete information and the other assuming none, and reconciling them requires recognising that the conflict is temporal rather than substantive. The set of boxes destined for a given charter is known days in advance: orders are assigned to post-harvest facilities four days before the warehouse date for Miami, and the air waybill exists before the truck departs the farm. What is unknown at execution time is not which boxes exist but in which order they will reach the conveyor. The problem therefore admits two distinct information horizons. At planning time the manifest is complete and an exact model can reason about weight, balance and capacity across the whole aircraft. At execution time only a short window of arrivals is visible and placement must be made irrevocably. Once this distinction is made explicit, the offline model and the online model are no longer competitors but consecutive stages, and the online problem is confined within each pallet rather than spanning the whole aircraft.

This study is guided by three research questions:

1. Does an upstream allocation stage that reasons about weight, balance and mix composition improve the achievable volumetric utilisation of an online packing procedure, relative to applying that procedure directly to the unstructured arrival stream?
2. How should the two stages be coupled, given that volumetric feasibility at the allocation level does not guarantee geometric packability under an online policy?
3. What is the marginal value, in utilisation and cost, of increasing control over the order in which boxes are released to the build-up conveyor?

The contributions are as follows. *Decomposition by information horizon.* We formalise the APP for full charter operations as a two-stage problem in which the upper stage is offline and exact and the lower stage is online and heuristic, in contrast to existing decompositions of the Air Cargo Load Planning Problem in which the packing subproblem is solved offline with complete information. *Mix composition as a decision variable.* The allocation stage does not only respect capacity; it can actively shape the dimensional composition of each pallet’s manifest, on the hypothesis that online packing performance depends on the heterogeneity of the stream and not only on its aggregate volume. *Weight and balance restored to an online formulation.* The allocation stage enforces a partial lateral balance proxy — a bound on the aggregate left–right weight difference over the side-by-side main-deck positions — together with payload and per-position weight limits, all absent from the online precursor study, so the packing stage can stay purely geometric without the load plan becoming inadmissible. Longitudinal moments, the certified centre

of gravity envelope and ULD tare stay outside the model. *Quantification of upstream coordination value.* By parameterising the degree of sequence control between full randomness and full determination, we produce an explicit curve relating coordination effort at the post-harvest facilities to utilisation gains at the terminal. *Industry-specific validation.* The model is validated on historical charter data for a commodity, cut flowers, that is largely absent from the air cargo palletisation literature despite its economic weight in the Latin American export basket.

The remainder of the paper is organised as follows. Section 2 describes the case study context and the operational process at the origin terminal. Section 3 reviews the literature on the Air Cargo Load Planning Problem, offline exact packing formulations and online packing policies, and positions the contribution. Section 4 formalises the two-stage problem. Section 5 details the solution methodology and the coupling mechanism. Section 6 presents the data and implementation. Section 7 reports and discusses results. Section 8 concludes.

2 Case Study Context

The sponsor company operates full charter flights from Bogotá’s El Dorado International Airport to Miami during demand peaks (Valentine’s Day, Mother’s Day), contracting an Airbus A330-200F at a fixed price per rotation. The aircraft configuration comprises 32 pallet positions: 21 on the main deck (16 side-by-side positions arranged in left–right pairs, one contoured side position, and four centre positions of types SS, TT, UU and VV) and 11 on the lower deck (one P9 position and ten LD positions). Each position has an individual volumetric capacity — already net of fuselage contour — an individual structural weight limit, and a position-specific weight utilisation coefficient between 0.5 and 1.0 reflecting the admissible share of the structural limit at that station. The aircraft-level payload limit for these operations is 65,000 kg, and ramp practice requires the lateral imbalance between left and right main deck positions to stay within 500 kg.

A representative charter manifest provided by the sponsor comprises 8,500 boxes of 35 commercial configurations (36 raw labels, one of which is a data-entry duplicate), with individual dimensions from $61 \times 13.5 \times 10$ cm to $107 \times 55 \times 46$ cm, weights from 2.9 to 18.8 kg, and a strong dominance of low, flat formats: half the boxes are at most 18 cm tall. Grouping boxes with identical dimensions, volume and weight yields 45 interchangeable classes, a structural property exploited throughout this study. At the terminal, boxes reach the build-up conveyor in an order determined upstream by truck arrival and unloading sequence; airline personnel stack them manually onto the open pallets (Figure 1), and a stacked box is not retrieved.



Figure 1: Build-up operation at the origin terminal during the authors' site visit: boxes of heterogeneous formats arrive on the conveyor and are stacked manually onto the open pallet positions under a first-come, first-loaded discipline.

3 Literature Review

This section reviews the industrial context, the canonical decomposition of air cargo load planning, the offline exact literature that dominates the palletisation subproblem, and the online packing literature that reflects terminal reality. It closes by identifying the gap that motivates the two-stage formulation.

3.1 The cut flower supply chain and its dependence on air freight

The economic structure of the cut flower trade explains why palletisation efficiency matters more here than in most air cargo segments. Floriculture in the Andean region is export oriented almost by definition, with negligible domestic demand relative to production volume, and production has shifted from traditional centres towards new producers, among them China, Colombia and Ecuador (Darras, 2021, p. 1). Latin American exporters compete on a combination of climatic advantage, proximity to the United States market and labour cost, but that advantage is mediated entirely by logistics performance (Vega, 2008a).

Perishability is the binding characteristic. Cut flowers and potted plants lose quality continuously as a function of time, temperature and humidity, so that supply chain design has to account for quality decay explicitly rather than check it at delivery (De Keizer et al., 2015, pp. 61, 74). Fresh food planning models therefore carry quality as a state of the product through the network and write off at a waste cost the consignments that fall below the required level (Rong et al., 2011, pp. 422–426). Network design for perishables trades transport cost and service time against losses from temperature and humidity exposure (Orjuela-Castro et al., 2021, p. 139), which favours short, reliable, temperature controlled paths. Between harvest and loading, flowers pass through grading, hydration, packing and refrigerated road transport, and their quality depends on the time and temperature of each step. A commercial trial on rose farms near Bogotá and Quito shows how far the effects of that first day travel: stems left the farm 25 hours after cutting and then spent up to eight days in transit to California at 2 to 17 °C, and handling decisions taken inside that first day were still measurable on arrival as up to 3.4 days of vase life and 2.5 times as many bruised petals (Macnish et al., 2009, pp. 216, 219–220).

Air freight is the only mode compatible with these requirements, and its cost weighs heavily on the exporter’s logistics account, with wide variation across origin countries attributable to route structure, competition among carriers and terminal efficiency (Button, 2020; Vega, 2008b). Comparative analysis of the Andean corridors confirms that the cost differential between exporting countries is driven less by distance than by the organisation of the freight chain: airfreight rates for perishables from South America to Miami track proximity only loosely — Ecuador paid US\$1.18 per kilogram in 2006 against US\$1.12 from the more distant Peru and US\$0.92 from Colombia — a pattern attributed to hub-

and-spoke structure, load factors, aircraft size, directional trade imbalance and airport capability rather than to distance (Vega, 2008b, pp. 326–328) (Vega, 2008a; Fioravanti et al., 2022). Global air cargo volume forecasts continue to project growth, which sustains rather than relieves the capacity pressure faced by seasonal shippers (IATA, 2025). Accounts of the industry emphasise that the reliability of air connections is itself a risk factor for exporters far from their markets (Van Liemt, 1999), and the innovation agenda for the Brazilian flower chain links post-harvest and transport technologies to export competitiveness (Paiva et al., 2024).

The sustainability argument closes the loop. Airline carbon emissions are proportional to fuel burn (Brueckner and Abreu, 2017, pp. 11, 15), so the question is how fuel burn responds to how full the aircraft is, and the econometric evidence does not allow it to be treated as fixed: in a panel of sixteen US passenger carriers over 1995–2015, a five percentage point rise in the weight-based load factor raises total fuel usage by 8.2% at fixed available tonne miles (Brueckner and Abreu, 2017, pp. 14–16). That estimate comes from passenger airlines, and in it the load factor is a commercial choice co-varying with network structure, not an experiment in which extra boxes are added to an identical flight. In a full charter the aircraft, the route and the schedule are contracted in advance, so the comparison is between flying the same flight fuller and operating additional flights: more payload per charter means fewer charters for a given seasonal volume, and it is the emissions of the avoided flights that are saved. Operator reporting uses emissions per available and per freight tonne kilometre (Baxter, 2020, pp. 204, 210–211); only the latter reflects utilisation. Comparative sector studies frame the problem in the same terms: Queiroz Júnior et al. (2024, pp. 874–876) benchmark twenty-one national aviation sectors by data envelopment analysis with carbon dioxide as an undesirable output, so that environmental performance is measured relative to the traffic produced rather than in absolute terms. Utilisation improvement is therefore a rare case in which the cost objective and the emissions objective are aligned rather than traded off.

Nota. [RESUELTA — puede borrarse antes de someter.] Tres referencias de esta subsección requieren atención antes de someter: (a) Sirisaranlak (2017) es un congreso no indexado que sostiene afirmaciones cuantitativas — verificar o sustituir; (b) Fioravanti et al. (2022) necesita el número de publicación del BID; (c) las dos referencias de Vega quedan desambiguadas como 2008a (working paper BID, DOI 10.18235/0011060) y 2008b (JATM, DOI 10.1016/j.jairtraman.2008.08.006). **Estado al 22/09/2026 (Daynor).** (a) Resuelto: la cuantificación “se mide en horas” se retiró y la oración se apoya ahora en De Keizer y en Macnish et al. (2009), rosas de Bogotá y Quito con DOI; la entrada de Sirisaranlak se eliminó de la bibliografía. (b) Resuelto: BID, DOI 10.18235/0004583. (c) Confirmado y ya aplicado en el .bib común.

3.2 The Air Cargo Load Planning Problem and its decomposition

Air cargo operations involve airlines, freight forwarders, terminal operators and ground transport providers, whose decisions interact across capacity planning, scheduling and routing; effective execution depends on coordinating constraints that belong to different organisations (Feng et al., 2015). Within this landscape, the loading of a single flight has been formalised as the Air Cargo Load Planning Problem (ACLPP), which Brandt and Nickel (2019) consolidate into four interdependent subproblems: the Aircraft Configuration Problem, which selects the types and number of unit load devices (ULDs) to build; the Build-up Scheduling Problem, which determines when and where each ULD is assembled; the Air Cargo Palletisation Problem (APP), which decides how items are packed within each ULD; and the Weight and Balance Problem (WBP), which positions the built ULDs in the hold so that the centre of gravity remains within the admissible envelope.

This taxonomy is the organising device for the present study, because it identifies precisely where the two precursor models act and why they are complementary rather than competing. The integer programming model addresses the APP at the level of assignment, deciding the composition of each ULD, and simultaneously enforces a subset of the WBP by constraining the lateral distribution of mass. The online packing model addresses the APP at the level of geometry, deciding the position and orientation of each box within a ULD whose contents have already been determined. Stated in the ACLPP vocabulary, the contribution of this paper is a coupled treatment of the assignment and geometric layers of the APP under a WBP constraint, with the Aircraft Configuration Problem and the placement of ULDs in the hold treated as exogenous inputs fixed by the carrier.

Brandt and Nickel (2019, §7, p. 409) report that in practice the planning steps are processed strictly sequentially, which gives local optimisation only, and that once the number of ULDs is fixed the three dimensional packing is replaced by a one dimensional consideration of cargo volume and weight (Brandt and Nickel, 2019, §3.2, p. 402). That observation describes the sponsor company’s situation exactly, and it also describes the limitation of the integer programming precursor study, whose volumetric constraint implicitly assumed perfect packing density — an assumption this paper quantifies and corrects in Section 4.3.

3.3 Offline formulations of the packing subproblem

The geometric core of the APP is the three-dimensional bin packing problem (3D-BPP), situated within the standard typology of cutting and packing problems of Wäscher et al. (2007), where it is a three-dimensional single bin size bin packing problem (Wäscher et al., 2007, p. 1124); that review counts 59 three-dimensional papers against 172 in one

dimension and 150 in two (Wäscher et al., 2007, p. 1126). Early analysis of approximation guarantees for bin packing traces to Johnson (1973), and the first widely adopted heuristic for three-dimensional container loading filled the container in layers, prioritising items that are harder to accommodate (George and Robinson, 1980). Martello et al. (2000) developed an exact branch and bound algorithm for the strongly NP-hard 3D-BPP, together with new lower bounds and approximation algorithms. Bortfeldt and Wäscher (2013) review 163 container loading papers published between 1980 and 2011 and give the first categorisation of the constraints that separate academic formulations from industrial practice: weight limits, weight distribution, orientation, load bearing and stability (Bortfeldt and Wäscher, 2013, §§4.1–4.5, pp. 3–8). These constraints are rarely combined: 76 of the 163 papers address two or more at once (Bortfeldt and Wäscher, 2013, p. 9). In their typology our allocation stage is a Multiple Heterogeneous Large Object Placement Problem, a type they single out as “neglected almost completely” (Bortfeldt and Wäscher, 2013, p. 12).

Applied work in air cargo has concentrated on mixed integer formulations of this core. Paquay et al. (2016) develop a mixed integer program for the 3D-BPP derived from an air cargo application, incorporating non-overlap along each axis, admissible rotations and stability. Recognising the computational limits of exact solution at industrial scale, the same group subsequently proposed MIP-based constructive heuristics with transportation constraints (Paquay et al., 2018a) and a tailored two-phase constructive heuristic for the Multiple Bin Size variant, in which containers of different dimensions may be selected (Paquay et al., 2018b). Alonso et al. (2019) extend integer programming to multi-container loading with practical constraints, including weight distribution and centre of gravity, in a road transport setting rather than air cargo. Sheng et al. (2016) combine tree search for pallet arrangement in a single container with a greedy procedure that fills residual gaps using infill boxes, reporting mean filling rates of 91.9–93.0% on the BR benchmark classes. Chan et al. (2006) develop a two-phase decision support system for air-cargo pallet loading oriented to practitioner use.

Weight and balance has its own literature and its own justification. Davies and Bischoff (1999) showed that weight distribution can be reconciled with high space utilisation in a single container, but only if the packing algorithm is designed for it: their layer-and-block heuristic produces interchangeable sections whose permutation and reflection bring the centre of gravity to within about 6% of the mid-point of the container length, and the existence of such sections is a pre-requisite of the method (Davies and Bischoff, 1999, pp. 524–526). Balance is therefore not free of the packing decision, and the guarantee weakens when the container is only partly filled. The maximum weights and the admissible centre of gravity range are set for each aircraft by the manufacturer and its type certificate; their application to loading is described in the FAA weight and balance guidance (FAA, 2016, chs. 1 and 9). Zhao et al. (2023) integrate cargo-to-ULD assignment and weight

and balance in integer linear programs for a B777F; relative to a bi-objective model that keeps the two assignments as separate submodels, their integrated models reduce the centre of gravity deviation from the target while achieving a similar payload. Palletisation is represented by aggregate volume only, without three dimensional packing (Zhao et al., 2023, p. 15).

3.3.1 Alternative computational paradigms

A separate strand explores non-classical computation for the same problem. The Airbus Quantum Computing Challenge framed aircraft loading as a benchmark for emerging hardware (Traversa, 2019), and Sotelo et al. (2021) encoded the same Airbus challenge as a knapsack with a volume and a weight limit and solved it with a Variational Quantum Eigensolver on a noise-free simulator, over small random instances and without reporting running times (Sotelo et al., 2021, pp. 435–438). The placement subproblem that the challenge statement itself lists — where to put the containers, the centre of gravity, the fuselage shear limits — is set aside. The strand shows that the industry regards the problem as computationally binding, but it selects rather than places, it is offline and single-vehicle, and it carries no balance constraint; to the best of our knowledge no published loading study addresses the A330-200F.

3.4 Online packing and the terminal reality

Every formulation reviewed above assumes that the full item set is known before placement begins. This is the decisive divergence from the process observed at the sponsor company’s export terminal, where boxes arrive on a conveyor in an order determined upstream by truck arrival and unloading sequence, and where a box once stacked is not retrieved. Under these conditions an offline optimum is not merely expensive to compute; it is not implementable, because the information it requires does not exist at the moment of decision.

The online variant of the bin packing problem addresses exactly this setting. Constructive heuristics remain central to it because they are computationally trivial and adapt in real time; the Bottom-Left rule, whose efficient implementation is due to Chazelle (1983) following the original two-dimensional formulation of Baker et al. (1980), minimises the resulting stacking height before lateral extent, which keeps pieces low in the container. Learning based approaches have more recently been applied to the same problem: Zhao et al. (2021a) show that deep reinforcement learning can acquire effective online placement policies using a predicted feasibility mask that restricts placements to physically stable positions. Tsang et al. (2025b) extend this line to two concurrently open identical bins with bin replacement strategies. Their double DQN outperformed the bottom-left, best-volume-fit, best-short-side-fit and best-long-side-fit heuristics most clearly with short

lookahead ($k = 5, 10$), while the advantage narrowed or vanished at $k = 15$ (Tsang et al., 2025b, pp. 10–12). The accompanying open source implementation, DeepPack3D, provides both the reinforcement learning agent and four constructive placement policies with configurable lookahead. It must be trained separately for each lookahead value (Tsang et al., 2025a).

The limitation of this literature for present purposes is symmetric to that of the offline strand. Online formulations model the arrival stream as exogenous and typically consider a single bin, or a small number of identical bins replaced sequentially. They do not represent a finite heterogeneous set of containers whose contents are jointly constrained by an aircraft level weight and balance envelope, and they offer no mechanism for influencing the composition of the stream itself.

3.5 Online packing with lookahead

Between the purely online and the offline settings lies a family of semi-online models in which the algorithm receives limited extra information or flexibility: a buffer, a preview of upcoming items, or sorted input (Epstein and Kleiman, 2009; Epstein and Levin, 2026). Under request lookahead of size l the algorithm sees the next l unserved items, and when these may be processed in any order the lookahead is equivalent to a reordering buffer of size l (Dunke and Nickel, 2021). Its value is problem dependent: for best-fit bin packing with one item of lookahead the gain is small and vanishes as the sequence grows (Dunke and Nickel, 2021), and a single-item buffer improves the best achievable asymptotic competitive ratio of cardinality constrained bin packing without any constant-size buffer closing the gap to the offline optimum (Epstein and Levin, 2026). In three dimensional packing a buffer of two or three boxes raises space utilisation (Puche and Lee, 2022), and Peeters et al. (2022) report diminishing returns to the visible window in industrial poultry batching. Our placement stage belongs to this class: the operator sees a window of $k = 7$ boxes and may place any of them. Controlling the order in which boxes are released upstream is a different lever, akin to the sorted-input model, and it is the one RQ3 examines.

3.6 Decision support and feedback loops

Decision support for air cargo loading has developed along two lines. At the ULD level, Lee et al. (2021) report a system, designed with a large cargo airline, that builds a single pallet with a genetic algorithm under aviation-safety and stability constraints and lets a supervisor trigger recalculation or relax tolerances; it attains a volumetric load factor around 70% against a practical target of 75% (Lee et al., 2021, pp. 1407, 1413). At the terminal level, Wong et al. (2021) embed a mixed-integer load-planning

model for a Boeing 747-8F, under compartment weight, centre-of-gravity and dangerous-goods constraints, inside a digital twin: sensors at gate-in measure the actual contour and weight of each consignment and the model is re-solved whenever these deviate from the plan (Wong et al., 2021, pp. 805–810). Closer to the present architecture, Chan et al. (2006) return to their aggregate pallet selection model when the geometric heuristic fails to load all items. The same group has since concentrated on the cost of the stability check itself, accelerating physical simulation on graphics hardware so that it can run inside the search rather than beside it (Mazur et al., 2022). Our coupling closes a loop of the same family, but it measures the geometric utilisation attained under randomised arrival orders rather than a physical reading, returns a calibrated position-specific volumetric capacity rather than a new plan, and feeds a more aggregate model rather than re-executing the same one. Its effect is also quantified here, which Wong et al. (2021) do not do (Wong et al., 2021, Table 2, p. 807).

3.7 Weight and balance as a separate exact layer

Exact models of the weight and balance problem take built ULDs, with their measured gross weights, and assign them to hold positions. Limbourg et al. (2012) load a given set of ULDs on a Boeing 747 under centre of gravity, lateral imbalance and cumulative load limits, minimising the moment of inertia. Closest to our assignment stage is Vancroonenburg et al. (2014), who select the most profitable subset of ULDs and position them so that weight and longitudinal centre of gravity stay within limits at zero-fuel, taxi, take-off and landing weights, under zone-wise left–right imbalance, cumulative, linear and floor constraints, solving 51 real flights in under ten seconds. Zhao et al. (2021b) replace tolerance bands by the piecewise-linear envelope and make the admissible lateral imbalance weight dependent. Metaheuristic treatments are coarser: Dahmani and Krichen (2013) assign items described only by weight and priority to identical containers and those to the six compartments of a Hercules C-130, under a single longitudinal moment constraint and with the centre of gravity computed as if every container sat at the geometric centre of its compartment (Dahmani and Krichen, 2013, pp. 2–3). Our assignment stage operates one level lower, deciding box-to-position assignments under volumetric and per-position weight capacities, so it represents only a partial lateral proxy of this layer. Longitudinal moments, the certified envelope, deck and zone load limits and ULD tare are left to the carrier’s weight and balance process (cf. the two-step commercial and loadmaster process in Limbourg et al., 2012, p. 4).

3.8 Perishables, and the gap addressed by this study

Optimisation for perishable goods in transport has concentrated on routing rather than packing. Hu et al. (2018) integrate inventory routing and freight consolidation for perish-

ables, minimising total transportation and inventory cost subject to a fixed shelf life after which a consignment must leave the network, and [Nguyen et al. \(2013\)](#) evaluate transportation practices in the California cut flower industry, examining consolidation centre design. Neither addresses the palletisation layer. Conversely, the air cargo palletisation literature reviewed above is largely commodity agnostic, and the specific characteristics of cut flowers — low density, high dimensional heterogeneity, fragility and cold chain dependence — appear in it only incidentally.

It is important to be precise about what is and is not novel here, because hierarchical treatments of the ACLPP already exist. [Brandt \(2017\)](#) decomposes the full load planning problem into its four subproblems and solves them mainly with mixed integer programming; for palletisation he uses a logic-based Benders decomposition coupling an item-to-ULD assignment master to a constraint programming packing subproblem, adding cuts and an estimated stowage loss when a ULD cannot be packed ([Brandt, 2017](#), pp. 98, 108). An earlier scheme instead caps ULD volume utilisation and re-solves the master with a reduced cap when a ULD proves infeasible ([Brandt and Nickel, 2019](#), p. 405). The architectural idea of an assignment master problem coupled to a packing feasibility subproblem is therefore established, and this paper does not claim it.

What has not been done, to the best of our knowledge, is to make that lower level subproblem *online*. In every existing decomposition the packing check has complete information about the items assigned to a container and may place them in any order. When the physical process forbids reordering and restricts visibility to a short lookahead window, three consequences follow that the existing literature does not address. First, feasibility becomes stochastic rather than deterministic, since a given assignment may be packable under one arrival permutation and not under another, which changes the nature of the cuts that can be generated. Second, the composition of the assigned set acquires an importance it does not have offline: a dimensionally homogeneous manifest is substantially easier for a myopic policy to pack than a heterogeneous one of identical total volume, which turns mix curation into a legitimate objective of the upper level model rather than a side effect. Third, the arrival sequence itself becomes a candidate decision variable, connecting the terminal problem to upstream coordination with the post-harvest facilities, a link that the sponsor company’s own operations have identified as a priority but that has not been quantified.

Table 1 positions the principal references against the criteria relevant to this study, including the two precursor models as separate rows. No single row satisfies all criteria, and in particular no row combines an offline exact treatment of allocation and balance with an online treatment of geometric placement: offline exact allocation with weight and balance and a geometric feasibility check does exist ([Brandt, 2017](#); [Puttaert, 2024](#)), but always with the packing subproblem solved offline. That combination is the contribution of this paper.

Table 1: Comparison of principal references against the criteria of this study.

Reference	Online	Multi-ULD	Offline exact	Weight & balance	Industry specific	Heur.	RL
George and Robinson (1980)	no	no	no	no	no	yes	no
Martello et al. (2000)	no	no	yes	no	no	yes	no
Davies and Bischoff (1999)	no	no	no	partial ^f	no	yes	no
Chan et al. (2006)	no	yes	no	partial ^a	air cargo	yes	no
Sheng et al. (2016)	no	no	no	no	furniture	yes	no
Paquay et al. (2016)	no	yes	yes	partial	air cargo	no	no
Paquay et al. (2018b)	no	yes	no	partial	air cargo	yes	no
Brandt (2017)	no	yes	partial ^c	yes	air cargo	yes	no
Alonso et al. (2019)	no	yes	yes	yes	road	no	no
Zhao et al. (2021a)	yes	partial ^d	no	no	no	no	yes
Zhao et al. (2023)	no	yes	yes	yes	air cargo	no	no
Tsang et al. (2025b)	yes	partial ^b	no	no	no	yes	yes
Precursor I (ILP allocation)	no	yes	yes	partial ^e	cut flowers	no	no
Precursor II (online packing)	yes	yes	no	no	cut flowers	yes	no
This study	yes	yes	yes	partial ^e	cut flowers	yes	yes

^a Pallet weight limits (pivot and maximum) and a heavier-at-the-base rule within each pallet; no aircraft centre of gravity and no position assignment.

^b Two identical bins open concurrently, replaced when full; unlimited supply rather than a finite heterogeneous set.

^c Logic-based Benders decomposition with a mixed integer master and a constraint programming packing subproblem; optimality is not guaranteed. Brandt and Nickel (2019) is a literature review and is therefore not listed as a method.

^d Multi-bin extension with identical bins.

^e Aggregate left-right weight difference over the side-by-side main-deck positions, plus payload and per-position weight limits; no longitudinal moment, certified envelope, deck limit or ULD tare.

^f Single container throughout (Davies and Bischoff, 1999, p. 510); the balance criterion is the distance of the centre of gravity from the geometric mid-point in length and width, as a soft post-processing objective. No payload limit, no vertical centre of gravity and no certified envelope (Davies and Bischoff, 1999, p. 527).

Nota. Los dos estudios precursores son un capstone GCLOG del MIT (equipo de autores distinto) y el manuscrito propio del B747. Antes de someter: o ambos son citables, o se reformulan en la Sección 2 como análisis internos antecedentes realizados con el patrocinador, acordándolo con Mejia-Argueta y Gutierrez-Franco.

4 Problem Definition

The two-stage problem is defined on a single charter flight. Let $\mathcal{I}$ be the set of boxes on the manifest ($N = |\mathcal{I}| = 8,500$ in the case study) and $\mathcal{J}$ the set of pallet positions ($M = |\mathcal{J}| = 32$). Each box i has real dimensions, volume v_i and gross weight m_i ; each position j has a volumetric capacity V_j (net of fuselage contour), a structural weight limit C_j^w , a weight utilisation coefficient $\lambda_j \in [0.5, 1]$, and a lateral tag in $\{L, R, C\}$; $\mathcal{J}_L, \mathcal{J}_R \subset \mathcal{J}$ denote the left- and right-tagged main-deck positions. The aircraft payload limit is $P = 65,000$ kg and the admissible lateral imbalance is $\Delta = 500$ kg.

4.1 Stage 1: allocation (offline, exact)

Stage 1 decides which boxes fly and on which pallet, with complete information. Binary variables x_{ij} assign box i to position j , u_i marks box i as left behind, and y_j activates position j . With penalty $\rho = 10$ per unassigned box:

$$\max \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} (m_i + v_i) x_{ij} - \rho \sum_{i \in \mathcal{I}} u_i \quad (1)$$

$$\text{s.t.} \quad \sum_{j \in \mathcal{J}} x_{ij} + u_i = 1 \quad \forall i \in \mathcal{I} \quad (2)$$

$$\sum_{i \in \mathcal{I}} v_i x_{ij} \leq V_j y_j \quad \forall j \in \mathcal{J} \quad (3)$$

$$\sum_{i \in \mathcal{I}} m_i x_{ij} \leq C_j^w \lambda_j y_j \quad \forall j \in \mathcal{J} \quad (4)$$

$$\sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} m_i x_{ij} \leq P \quad (5)$$

$$\left| \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}_L} m_i x_{ij} - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}_R} m_i x_{ij} \right| \leq \Delta \quad (6)$$

$$x_{ij}, y_j, u_i \in \{0, 1\},$$

where (6) is linearised with an auxiliary continuous variable $d \geq 0$ in the standard way. This is the formulation of precursor study I (variant W, V, P, selected there by sensitivity analysis), which we replicate and verify in Section 7.1.

Three properties of (1) deserve explicit statement. First, the objective mixes kilograms ($m_i \in [2.9, 18.8]$) and cubic metres ($v_i \in [0.02, 0.11]$): the volume term is two

orders of magnitude smaller and acts as a tie-breaker, so the objective is in effect a lexicographic weight-first maximisation. Second, by (2) the penalty term equals $\rho(N - \sum_{ij} x_{ij})$ up to a constant, so ρ is a uniform bonus per assigned box that favours placing more (lighter) boxes among weight-equivalent solutions. Third, because thousands of boxes are identical, the model is massively symmetric: the optimal totals are stable but the per-pallet composition is not unique. Boxes with identical dimensions, volume and weight are interchangeable, which permits an aggregated reformulation over the 45 box classes with integer variables X_{tj} that attains the same optimum and solves in under a second (Section 7.1); all Stage-1 extensions in this paper operate on the aggregated form.

4.2 Stage 2: placement (online, geometric)

Stage 2 receives, for each position j , the manifest $\mathcal{B}_j = \{i : x_{ij} = 1\}$ selected by Stage 1, and must stack it physically. Boxes arrive in a sequence $\pi = (b_1, \dots, b_n)$ unknown at planning time; at step t only the lookahead window $\mathcal{W}_t = (b_t, \dots, b_{t+k-1})$ is visible ($k = 7$ throughout), and a placement, once made, is irreversible. The bin is the rectangular prism (W_j, H_j, D_j) of the position, discretised on a 1-cm grid with conservative rounding (box dimensions up, bin dimensions down; Section 6.2). A placement of box b with axis-aligned rotation r at origin (p^x, p^y, p^z) is feasible if it is contained in the bin, does not overlap previously placed boxes, and satisfies the gravitational stability requirement that strictly more than a fraction $\alpha = 0.5$ of its base footprint rests at the placement height on the current height map $\mathbf{H}_j$:

$$\frac{|\{(x, z) \in \text{footprint}(b) : \mathbf{H}_j(z, x) = p^y\}|}{w' d'} > \alpha, \quad p^y = \max_{(x, z) \in \text{footprint}(b)} \mathbf{H}_j(z, x). \quad (7)$$

The objective of Stage 2 is to maximise, for each position, the number of manifest boxes actually placed (equivalently, given box volumes, the geometric utilisation of the position); boxes that cannot be placed when their turn comes are diverted, and their count is the measure of the allocation–packability gap that couples the two stages.

4.3 Coupling

Volumetric feasibility in Stage 1 does not imply geometric packability in Stage 2: a manifest that fits a pallet’s volume V_j on paper may not be stackable by an online policy under any realistic arrival order. The coupling mechanism studied here replaces V_j in (3) with an *empirically calibrated fill factor*

$$\hat{V}_j = \min(V_j, \hat{u}_j \cdot V_j^{\text{prism}}), \quad (8)$$

where $\hat{u}_j$ is the geometric utilisation achieved by the online policy on position j in the previous iteration, measured only on positions whose packing was geometry-limited (positions that ran out of boxes retain their nominal capacity). Iterating Stage 1 and Stage 2 under (8) shifts allocation away from positions the online policy cannot fill, trading paper volume for physically realised load.

5 Solution Methodology

5.1 Two-stage decision support model

The DSM operates as follows. (1) Solve the Stage-1 ILP (aggregated form) with Gurobi to optimality tolerance 10^{-3} . (2) Materialise the aggregated solution into a box-level assignment (classes are interchangeable). (3) For each pallet position, simulate the on-line placement of its manifest under S random arrival permutations with each candidate policy, recording placed boxes, diverted boxes and geometric utilisation. (4) Optionally, recalibrate fill factors (8) and return to step 1. As a benchmark for RQ1, the online engine is also run *standalone*: the full 8,500-box stream is fed to all 32 open positions simultaneously with no Stage-1 guidance, replicating the single-stage architecture of precursor study II.

5.2 Space representation and placement policies

The packing engine is the DeepPack3D framework (Tsang et al., 2025a,b) as extended by precursor study II to heterogeneous bins. Free space in each bin is a dynamic set of maximal free cuboids; placing a box splits every intersecting free cuboid into up to six maximal sub-cuboids and discards dominated ones, while a height map $\mathbf{H}_j \in \mathbb{N}^{D_j \times W_j}$ records stacking heights. Four deterministic policies score the feasible placement set: Bottom-Left (BL), which minimises resulting height, then rightward and forward extent (Chazelle, 1983); Best Short Side Fit (BSSF), which minimises the smaller residual dimension of the hosting free cuboid; Best Area Fit (BAF), which prefers the smallest hosting cuboid with BSSF as tie-break; and Best Long Side Fit (BLSF), which minimises the larger residual (Tsang et al., 2025a,b). All four evaluate every feasible (box, rotation, position) triple in the lookahead window and select the minimum score.

5.3 Deep reinforcement learning agent

The learning policy is the DQN agent of DeepPack3D, an instance of deep Q-learning (Mnih et al., 2015): a convolutional Q-network over three 32×32 spatial inputs (current height map, post-placement height map, constant plane) concatenated with an embedding of the remaining lookahead items, trained with replay memory, an ε -greedy behaviour

policy, Bellman targets $r + \gamma \max_a Q(s', a)$ with $\gamma = 0.95$, and a target network mixed 50/50 into the behaviour network every 10 gradient steps. The reward is the average of a *pyramid* term (placed volume over summed height map) and a *compactness* term (placed volume over the enclosing prism of the current stack).

Two implementation adaptations were required for the A330 setting, and both are documented as such. First, the published network expects 32×32 observations, while the cm-grid height maps of the A330 positions are up to 318×244 ; observations are therefore compressed by exact block-maximum pooling to 32×32 (the pooled post-placement map equals the pooled version of the full-resolution map, so no approximation is introduced beyond the fixed observation resolution). Second, height normalisation uses the height of the bin being evaluated rather than a global reference size, correcting a calibration inconsistency of the original package that is harmless with homogeneous bins but material with heterogeneous ones. The agent was reimplemented layer-for-layer in PyTorch/CUDA because TensorFlow no longer supports GPU execution on native Windows; architecture, hyperparameters, reward and training procedure follow the original. Training episodes present a single position (cycling through the 32 real positions) with a conveyor drawn from the empirical box distribution of the manifest; ε decays by factor 0.95 every 10 episodes to a floor of 0.025. Two further adaptations concern the replay buffer: transitions are stored as self-contained encoded states (the compressed observation plus bin metadata and integer action records), which bounds memory at industrial state sizes and ensures that transitions collected on one position are never re-evaluated against another position’s geometry; and the bootstrap maximum $\max_a Q(s', a)$ is computed over a fixed random subsample of at most 64 feasible actions of s' , recorded when the transition is stored.

5.4 Arrival sequence control

To answer RQ3, the arrival sequence is parameterised by a control level $c \in [0, 1]$ interpolating between the uncontrolled terminal ($c = 0$: uniformly random permutation) and fully coordinated release ($c = 1$: boxes sorted by decreasing footprint, then decreasing height, grouped by class — the order a base-first manual stacking doctrine would request). Intermediate values mix the ranks of the two orders, so c measures the fraction of ordering information the post-harvest facilities inject into the stream. For each c the pipeline is re-simulated with S seeds and the placement rate traced as a function of c .

5.5 Key performance indicators

For each configuration we report: boxes placed and placement rate (placed over assigned); geometric utilisation per position (placed discrete volume over position prism volume); realised load (kg and m^3 of the actually placed boxes, mapped through the interchangeable class of each placed size); and, for standalone runs, weight feasibility diagnostics (per-

position weight versus $C_j^w \lambda_j$, aircraft total versus P , lateral imbalance versus Δ) computed from class-average weights. Statistical comparisons between policies use the one-sided Wilcoxon signed-rank test (Wilcoxon, 1945) paired by arrival-sequence seed, following the protocol of precursor study II.

6 Case Study and Implementation

6.1 Data

The dataset is the historical charter manifest of precursor study I: 8,500 boxes over 35 cleaned commercial types (45 interchangeable dimension–weight classes; 29 distinct cm-grid dimension classes), and the 32-position A330-200F configuration with per-position volumetric capacities (16.74 m³ main deck side-by-side, 12.40 m³ lower deck LD, 7.80 m³ P9, 17.10 m³ contoured side position), structural weight limits (2,809–5,185 kg) and weight utilisation coefficients (0.5–1.0). Total nominal volumetric capacity is 482.0 m³ (net of contour); the summed prism volume of the positions is 519.6 m³.

6.2 Implementation

Stage 1 is implemented in Python 3.12 with Gurobi 13.0.2 (Gurobi Optimization, 2024); the box-level model has 272,000 binaries and solves to a 0.04% gap in 5 s, the aggregated model in under 0.1 s. An independent GAMS 49.4 implementation of both forms is provided for cross-solver validation; the box-level model reproduces the allocation on desktop GAMS in 21.7 s (Section 7.1). Stage 2 runs the DeepPack3D engine (Python, NumPy) on a 1-cm grid: box dimensions are rounded up to the next centimetre and bin dimensions truncated down, a conservative discretisation that inflates total box volume by only −0.4% (the small negative sign arising from the dataset’s own rounded volume column). The decimetre convention of precursor study II (`div10_ceil`) was evaluated and rejected for this dataset: with box heights of 10–20.5 cm, ceiling to decimetres inflates total box volume by +43.9%, which would make the Stage-1 allocation geometrically infeasible by construction. Where the precursor’s decimetre grid is used for comparison it is labelled explicitly. The DQN agent trains on an NVIDIA RTX 4060 (8 GB) with PyTorch 2.6/CUDA 12.4 (Paszke et al., 2019). Unless stated otherwise, results aggregate $S = 10$ random arrival seeds per configuration (five for the coupling and sequence experiments), with $k = 7$ and $\alpha = 0.5$.

7 Results and Discussion

7.1 Stage 1: replication and validation of the exact allocation

The replicated ILP reproduces the results of precursor study I within the optimality tolerance: 7,967 boxes assigned (precursor: 7,966), 64,986.2 kg (64,998.4), 392.61 m³ (392.64), objective 60,048.8 at a 0.039% gap, lateral imbalance 438.4 kg $\leq$ 500 kg. The aircraft payload limit (5) is binding — weight, not volume, is the scarce resource at the flight level — and 533 boxes (mostly the largest formats) remain unassigned. The aggregated 45-class reformulation attains the identical objective, box count and load totals in under a second, confirming the interchangeability argument of Section 4.1; per-pallet compositions differ across alternative optima, as expected under symmetry, so all downstream results are reported against totals and against a fixed representative assignment. Cross-solver validation on desktop GAMS 49.4 confirms the replication: the box-level model terminates normally within the same 10^{-3} tolerance at objective 60,062.1, assigning the identical 7,967 boxes (533 unassigned) with 64,999.4 kg and 392.69 m³, and a lateral imbalance of 120.5 kg — totals within 0.02% of the Gurobi incumbent, with the small differences again reflecting alternative optima inside the common optimality tolerance.

7.2 Stage 2 over Stage 1: the allocation–packability gap

Table 2 reports the placement of the Stage-1 manifest by each online policy over 10 random arrival sequences. The best policy, BL, physically places $7,041 \pm 53$ of the 7,967 assigned boxes ($88.38\% \pm 0.66$), followed by BSSF (84.50%), BAF (79.04%) and BLSF (71.52%); all pairwise differences from BL are significant (one-sided Wilcoxon paired by seed, $W = 55$, $p < 0.001$, $n = 10$). In physically realised load, BL places $57,023 \pm 421$ kg of the 64,986 kg allocated on paper, already +1.2% above the historical manual load of 56,332 kg before any coupling. The policy ranking replicates the finding of precursor study II on a different aircraft, dataset and grid, strengthening the external validity of BL as the operational recommendation.

The learning agent lands third. Trained for 640 single-position episodes on the empirical box distribution (Figure 2; episode utilisation rises from 0.526 to 0.652 as ε anneals to 0.042), the DQN places $6,521 \pm 40$ boxes (81.84%, geometric utilisation 0.623): significantly above BAF and BLSF, significantly below BL and BSSF (one-sided Wilcoxon vs BL, $p < 0.001$). This differs from Tsang et al. (2025b), whose double DQN beat the best constructive heuristic by about 5–6 points at $k = 5$ and $k = 10$ on synthetic cubic bins and lost that advantage only at $k = 15$. Our budget of 640 single-position episodes is about two orders of magnitude below the 10^5 episodes they use, and DeepPack3D ships no pretrained model for $k = 7$, so the finding fits their diagnosis of insufficient exploration rather than contradicting their result. Learning-based packers are trained at a far larger

scale, for hours or tens of millions of environment steps, and attention-based policies remove the dependence of the action space on container size that here forced the observation compression (Zhao et al., 2021a; Xiong et al., 2024; Zhang et al., 2021); we did not evaluate them. The conclusion is narrow: a standard DQN at an affordable training budget does not beat bottom-left on this industrial instance, and the recommended policy needs no GPU, no training and no maintenance.

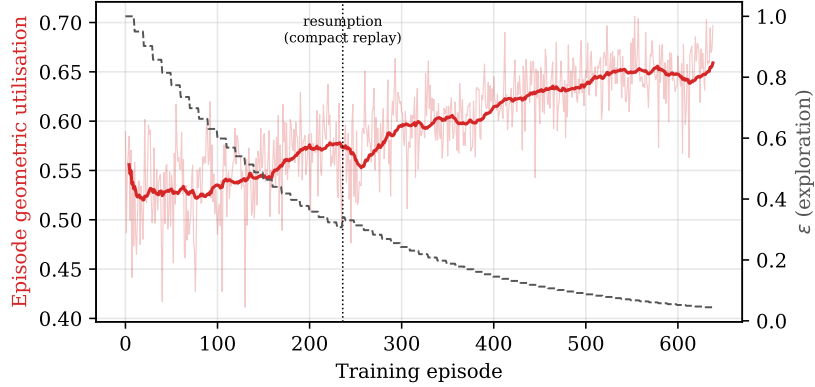


Figure 2: DQN training on single-position episodes drawn from the empirical box distribution: per-episode geometric utilisation (rolling mean) and exploration schedule. The dotted line marks the resumption under the compact replay encoding (Section 5.3).

The headline result is the gap itself: roughly *one in nine boxes that the exact allocation declares feasible cannot be stacked* by the best online policy under realistic arrival uncertainty. The gap concentrates in the flat main-deck positions (geometric utilisation 0.60–0.80) while the lower-deck LD containers pack at 0.83–0.90; positions whose manifests were weight-limited by Stage 1 (e.g. the 0.7-coefficient side positions) run out of boxes before running out of space. This is the quantitative content of RQ2: aggregate volumetric capacity is an optimistic proxy for online packability, and the error is position-dependent.

Table 2: Placement performance of the online policies over the Stage-1 allocation (10 random arrival sequences, $k = 7$, cm grid). Rate = placed/assigned. p -values: one-sided Wilcoxon signed-rank versus BL, paired by seed.

Policy	Boxes placed	Rate (%)	Geom. util. (%)	p vs BL
BL	7041 $\pm$ 53	88.38	68.33 $\pm$ 0.48	—
BSSF	6733 $\pm$ 43	84.50	64.22 $\pm$ 0.54	0.00098 ***
BAF	6297 $\pm$ 35	79.04	60.27 $\pm$ 0.40	0.00098 ***
BLSF	5698 $\pm$ 54	71.52	54.38 $\pm$ 0.58	0.00098 ***
DQN	6521 $\pm$ 40	81.84	62.29 $\pm$ 0.41	0.00098 ***

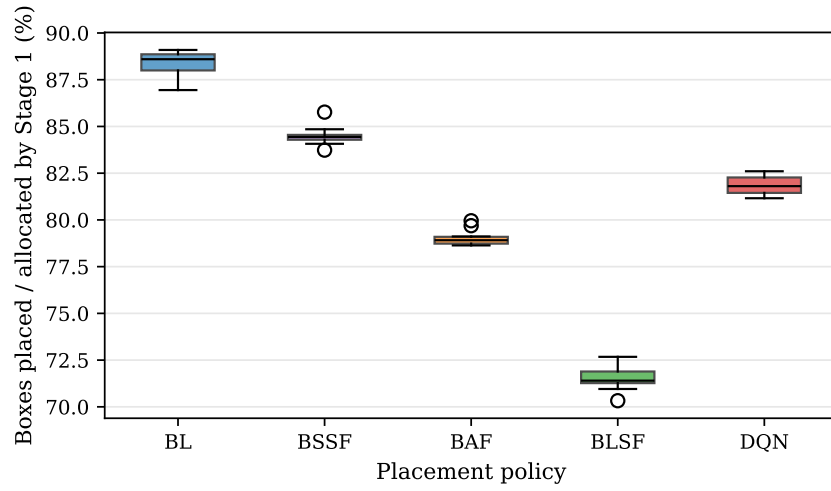


Figure 3: Distribution over arrival seeds of the placement rate of each online policy applied to the Stage-1 allocation.

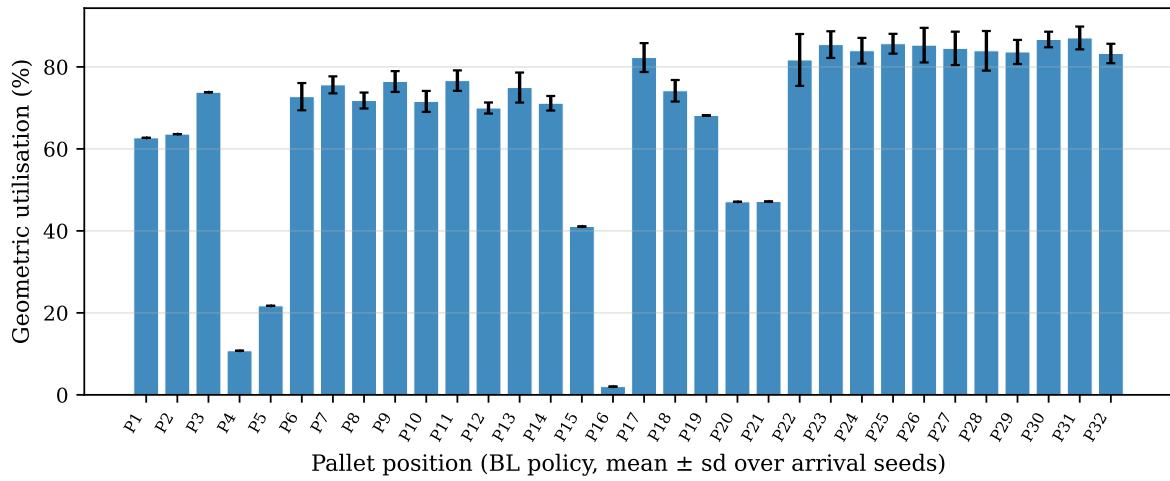


Figure 4: Geometric utilisation by pallet position under the best policy (BL), mean $\pm$ sd over arrival seeds.

7.3 The value of the allocation stage (RQ1)

The single-stage benchmark feeds the full 8,500-box stream to all 32 open positions with no upstream allocation, replicating the architecture of precursor study II. Over 15 arrival seeds, standalone BL places $7,944 \pm 50$ boxes (93.5% of the manifest) at a mean geometric utilisation of 0.798 ± 0.005 ; BSSF, BAF and BLSF follow at 0.726, 0.680 and 0.602 ($n = 10$ each). On geometry alone, the single-stage packer therefore *outperforms* the two-stage pipeline (Figure 5): it sees every box, not only the weight-feasible subset selected by Stage 1, and it fills the prisms densely.

That comparison is precisely the trap the two-stage architecture is designed to avoid, because the standalone plan is not operationally admissible. Reconstructing the weight of each standalone solution from the class-average weight of every placed box shows that the dense geometric packings systematically violate the constraints that Stage 1 enforces by construction: the standalone BL load averages 68,583 kg over six audited seeds, exceeding the 65,000 kg aircraft payload limit in *every* run, with 7.7 pallet positions above their structural weight limits on average (BSSF, BAF and BLSF: 6.0, 4.3 and 2.0 positions over limit, $n = 10$ each). Lateral imbalance remains within the 500-kg bound in all runs — randomised arrival spreads mass roughly symmetrically — but per-position and aircraft-level weight admissibility fails wherever the packer is dense enough to matter, and it fails worst for the best geometric policy.

The answer to RQ1 is therefore conditional in an instructive way. An upstream allocation stage does not increase raw geometric utilisation — it cannot, since it restricts each position’s manifest — but it is what converts geometric density into an *admissible* load plan: the coupled pipeline of Section 7.4 realises 63,346 kg within every weight, balance and capacity bound, whereas the standalone packer’s nominally larger load could not legally depart. Single-stage online packing at aircraft scale is a useful bound, not an operating procedure.

7.4 Coupling by calibrated fill factors (RQ2)

Replacing the nominal volumetric capacities of Stage 1 with the fill factors (8) calibrated from the previous iteration’s measured utilisation has a large effect (Figure 6). One calibration step raises the placement rate of the same 7,967-box, 65,000-kg allocation from $88.5\% \pm 0.5$ to $97.3\% \pm 0.3$, and the physically realised load from $57,135 \pm 310$ kg to $63,346 \pm 167$ kg (the coupling and sequence experiments use five dedicated seeds; their iteration-0 baselines agree with the ten-seed figures of Table 2 within one standard deviation) — an additional 6.2 tonnes per flight obtained purely by redistributing the same assigned load towards positions the online policy can actually fill. Against the sponsor’s historical manual load of 56,332 kg, the coupled two-stage model realises +12.5% more cargo on the same aircraft.

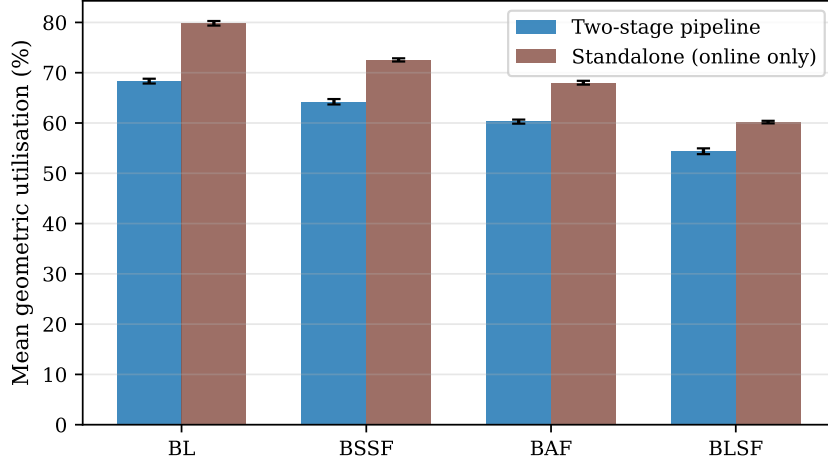


Figure 5: Mean geometric utilisation, two-stage pipeline versus standalone single-stage packing (10 arrival seeds per bar). The standalone bars are geometrically denser but correspond to weight-inadmissible plans (Section 7.3).

Two properties of the iteration deserve note. First, naïve recalibration oscillates: at iteration 2 the placement rate falls back to 91.8%, because the improved placement leaves few geometry-limited positions, their caps revert to nominal, and the allocation drifts back into the unpackable regime. A monotone update, in which caps never re-loosen ($\hat{V}_j^{(t+1)} = \min(\hat{V}_j^{(t)}, \hat{u}_j V_j^{\text{prism}})$), removes the oscillation: rates progress 88.5% $\rightarrow$ 97.3% $\rightarrow$ 96.9% $\rightarrow$ 98.8% across iterations, but assigned boxes and realised load peak at the first calibrated iteration (7,967 boxes, 63,346 kg) and then decline gently (7,822 boxes, 62,218 kg at iteration 3) as caps keep tightening — so a single calibration step captures essentially the whole coupling gain, and further iterations trade realised load for placement rate. Second, the calibrated MIP solution is markedly more homogeneous per position (2.6 box classes per pallet on average), which anticipates the mix-composition analysis of Section 7.6.

7.5 The marginal value of sequence control (RQ3)

Figure 7 traces placement performance as the arrival stream moves from fully random ($c = 0$) to fully coordinated release ($c = 1$). The curve is nearly flat: 88.9% $\pm$ 1.0 at $c = 0$ against 90.2% (deterministic) at $c = 1$, a maximum gain of 1.4 percentage points, with intermediate levels statistically indistinguishable from the uncontrolled baseline. Geometric utilisation rises slightly (0.706 $\rightarrow$ 0.727).

The operational reading answers RQ3 in a direction the sponsor did not anticipate: under a competent allocation stage and a modest lookahead window ($k = 7$), the online policy already absorbs most of the sequencing noise, and investing in upstream release coordination buys less than investing in allocation quality (+1.4 pp against +8.8 pp for fill-factor coupling). Coordination effort at the post-harvest facilities is therefore better

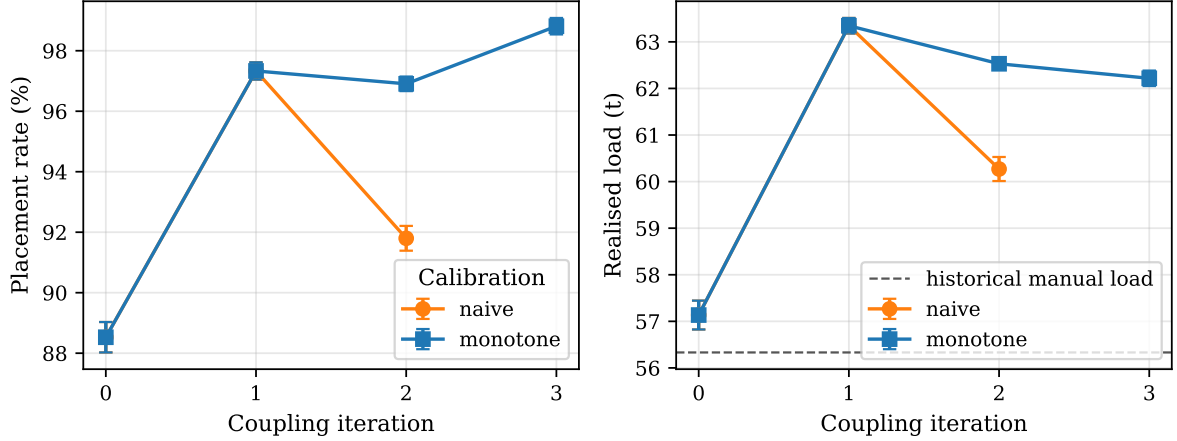


Figure 6: Fill-factor coupling across iterations: placement rate (left) and physically realised load (right); dashed line marks the historical manual load. Naïve recalibration oscillates; the monotone rule converges.

spent guaranteeing manifest accuracy than enforcing arrival order.

This figure is a lower bound on the value of sequence control. The coordinated level $c = 1$ is a single heuristic order (decreasing footprint, decreasing height, grouped by class), and even the classical decreasing pre-order carries a worst-case guarantee strictly above the optimum (Dunke and Nickel, 2021, p. 206), so it cannot be the best attainable order. Semi-online analysis separates a reordering buffer from a sorted input (Epstein and Levin, 2026; Epstein and Kleiman, 2009), whereas our design varies both at once. The experiment establishes that this order, under BL with $k = 7$ and this manifest, gains 1.4 percentage points over random arrival. A flat curve is also not proof that the information is worthless: in industrial batching the value of a wider window depends on the algorithm’s ability to exploit it (Peeters et al., 2022, p. 12), so a myopic rule such as bottom-left may leave on the table what a stronger policy would capture.

7.6 Mix composition (contribution 2)

To test the hypothesis that online packability depends on the dimensional heterogeneity of each pallet’s manifest and not only on its aggregate volume, the aggregated Stage-1 model is augmented with a penalty $\mu \sum_{t,j} z_{tj}$ on the number of distinct box classes present in each position (binary indicators z_{tj} linked by $X_{tj} \leq n_t z_{tj}$, with n_t the number of boxes of class t), and μ is swept while holding all capacity constraints fixed (Table 3). The effect is unambiguous: raising μ from 0 to 20 reduces the mean number of classes per pallet from 1.91 to 1.50 and raises the BL placement rate from 92.1% to 97.2%, at essentially zero cost in allocated boxes or weight (7,961 vs 7,967 boxes; 64,994 vs 64,986 kg). Homogeneity is close to a free good at the allocation level, and it buys

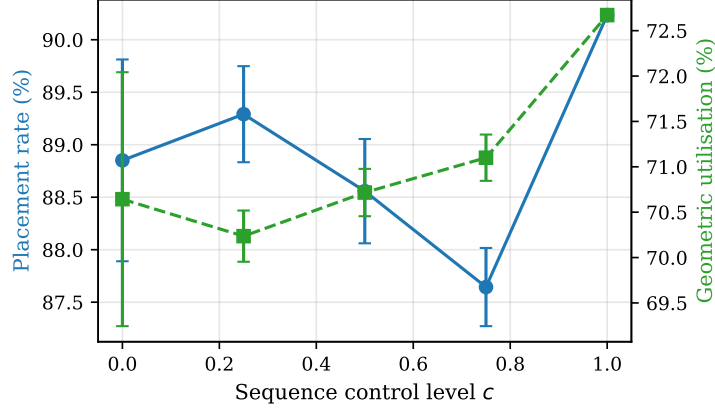


Figure 7: Placement rate and geometric utilisation as a function of the arrival-sequence control level c (BL, 5 seeds per level).

placement performance comparable to fill-factor coupling through an entirely different lever — composition rather than capacity.

The sweep also exposes a subtler phenomenon. The $\mu = 0$ aggregated baseline already places at 92.1%, nearly four percentage points above the box-level Stage-1 solution of Section 7.2 (88.4%) — yet both are alternative optima of the same ILP within the solver tolerance. Because thousands of boxes are interchangeable, Stage 1 admits a large optimal face: solutions indistinguishable in objective differ by almost 4 percentage points in online packability purely through how classes happen to be distributed, and admitting near-optimal solutions (the $\mu = 20$ allocation sacrifices $\approx 0.1\%$ of the original objective) extends the spread to almost 9 points. This is direct evidence for the second research question: *which* optimal allocation is chosen matters operationally, and the choice should be steered by packability-aware terms such as μ or the fill factors of Section 7.4.

Table 3: Mix-composition sweep: penalising dimensional heterogeneity in Stage 1 (aggregated model) and packing the result with BL (5 seeds).

μ	Classes/pallet	Boxes assigned	kg assigned	Placement rate (%)	Geom. util.
0	1.91	7967	64,986	92.07	0.68
1	1.84	7967	64,994	93.83	0.70
5	1.78	7967	64,986	94.25	0.71
20	1.50	7961	64,994	97.18	0.72

7.7 Financial and environmental impact

The economic reading follows the cost model of precursor study I: the charter is contracted at US\$59,697 per flight, so unit freight cost is the charter price divided by the kilograms actually flown. The historical manual operation loads 56,332 kg (US\$1.060 per kg). Precursor study I reported that its exact allocation would fly 64,998 kg, implying US\$0.918

per kg, a 13.3% unit cost reduction — but that figure rests on allocated, not stackable, load. Passing the allocation through the online placement stage shows what survives contact with the conveyor: the uncoupled pipeline realises $57,023 \pm 421$ kg (US\$1.047 per kg, -1.2%), and the fill-factor-coupled pipeline realises $63,346 \pm 167$ kg, i.e. US\$0.942 per kg and a defensible **11.1%** unit cost reduction against the manual baseline. The 2.2-point spread between the paper claim (13.3%) and the realised claim (11.1%) is precisely the quantity that single-stage studies cannot observe, and reporting it is a contribution of the two-stage design. At the sponsor’s twenty peak-season charters per year, the realised gain of 7,014 kg per flight amounts to roughly 140 tonnes of additional export capacity annually on unchanged flying.

The environmental reading follows, with one qualification. Emissions are proportional to fuel burn (Brueckner and Abreu, 2017, p. 11), and freighter operators account for performance per freight tonne kilometre (Baxter, 2020, pp. 210–211). At fixed seasonal volume, the +12.5% in payload per charter (56,332 to 63,346 kg) moves the same tonnage in 11.1% fewer flights, and the emissions of those flights are avoided in full. The two percentages coincide by construction: both the unit cost and the number of flights required scale with the inverse of the payload actually flown. Per flight, the fall in carbon intensity per kilogram is bounded above by 11.1% and is smaller in practice, since more weight raises fuel burn: the only large-sample estimate we are aware of gives +8.2% in fuel per five percentage points of weight-based load factor (Brueckner and Abreu, 2017, pp. 15–16). We report the avoided-flight figure because it does not require transferring a passenger airline elasticity to a freighter.

Nota. Decisión de los cuatro autores: el argumento ambiental pasó de “menos emisiones por kilo” a “menos vuelos a volumen estacional dado”. El motivo es que Brueckner y Abreu (2017) miden +8,2% de combustible por cada +5 puntos de factor de carga, es decir, el consumo NO es fijo respecto a la carga, y su muestra son aerolíneas de pasajeros de EE.UU., sin cargueros. La cifra de $-11,1\%$ por kilo queda como cota superior. Confirmar la aritmética de 9,9% contra las corridas definitivas. **Estado al 22/09/2026 (Daynor).** Aritmética verificada y **corregida:** con 56,332 kg en la operación manual y 63,346 kg en la tubería acoplada, la carga por vuelo sube 12,5% y los vuelos necesarios bajan 11,1%, no 9,9%. El 9,9% venía de tratar el 11,1% de ahorro unitario como si fuera el aumento de carga. El porcentaje de vuelos evitados coincide con el de ahorro unitario porque ambos van con el inverso de la carga volada. Sigue abierta la decisión de los cuatro autores sobre el encuadre del argumento.

8 Conclusions

This paper formalised the palletisation of a full charter flower flight as a two-stage problem separated by information horizon — an exact allocation stage that sees the complete manifest, and an online placement stage that sees only a seven-box window of an irreversible arrival stream — and validated the resulting decision support model end-to-end on a historical 8,500-box manifest of an Airbus A330-200F. Five conclusions follow.

First, the exact allocation stage of the precursor study replicates and is correct as stated, but its volumetric feasibility is an optimistic proxy for reality: roughly one in nine boxes it declares feasible cannot be stacked by the best online policy under realistic arrival uncertainty. Quantifying this allocation–packability gap at aircraft scale, and the mechanisms that close it, is the central contribution.

Second, among online placement policies, Bottom-Left dominates BSSF, BAF and BLSF significantly and consistently, replicating the finding of the precursor study on a different aircraft, dataset and grid — strong external validity for a policy simple enough to brief to ramp personnel.

Third, the two stages must be coupled, and one empirical calibration step suffices: feeding the measured achievable utilisation back into the allocation model as position-specific fill factors raises realised load from 57.0 to 63.3 tonnes per flight (+12.5% over the manual baseline; 11.1% lower unit freight cost), while a monotone update rule prevents the oscillation that naïve recalibration exhibits. Composition is a second, nearly free coupling lever: penalising dimensional heterogeneity within positions raises packability to 97.2% without measurable allocation cost — and the fact that online packability varies by nine percentage points *across alternative optima of the same ILP* shows that packability-aware selection among optimal allocations is not optional.

Fourth, the single-stage benchmark clarifies what each stage is for: pure online packing at aircraft scale is geometrically denser but produced weight-inadmissible plans in every audited replication. Allocation buys admissibility, not density; placement buys density within admissibility. By contrast, controlling the arrival sequence — the coordination lever the sponsor considered most promising — is worth at most 1.4 percentage points under a competent allocation, an order of magnitude less than allocation quality itself.

Fifth, methodologically, discretisation is not a detail: the decimetre convention inherited from the precursor inflates the volume of flat flower boxes by +43.9% and would have rendered the coupled analysis meaningless, whereas a 1-cm conservative grid is exact to −0.4%. Transfer of packing methodology between datasets must revalidate the grid against the item geometry.

Sixth, a standard DQN at a practically affordable training budget is not superior in this regime: the DQN agent, trained on the empirical box distribution and evaluated under the same protocol, places 81.8% of the allocation — above two of the four constructive

policies, below Bottom-Left by 6.5 percentage points with high significance. This runs against Tsang et al. (2025b), who report the opposite ranking at comparable lookahead with a training budget about two orders of magnitude larger. It still matters operationally: the best available policy is also the cheapest to run, explain and certify.

8.1 Limitations

The Stage-2 model treats each pallet as a rectangular prism, while the nominal capacities of Stage 1 already discount fuselage contour; the two geometries bracket the truth and their difference is reported explicitly. Weight is enforced at allocation, not during stacking, so within-pallet vertical weight distribution (heavy-below) is not modelled. Box weights within an interchangeable dimension class are averaged for the realised-load accounting. The DQN agent is trained and evaluated under the computational adaptations of Section 5.3, and its training budget (640 episodes) is modest by deep RL standards; the comparison bounds what a practically trainable agent achieves, not what RL can achieve in the limit. The weight-feasibility audit of the standalone benchmark uses class-average weights rather than box identities, and the fill-factor calibration is specific to the policy (BL) and lookahead ($k = 7$) with which it was measured. Stability is enforced by the partial base support rule of Section 4.2, with $\alpha = 0.5$. Benchmarked against a dynamic multibody simulation of the loading sequence, that family of geometric rules reproduces the simulated stability of a layout in 74.7–77.6% of cases, and the $\alpha = 0.5$ variant falls to 68.0% once centres of mass are displaced from the geometric centre of the box (Mazur et al., 2025, pp. 8–9). Cut flower boxes are that case. The error is two sided: underestimating stability costs utilisation, overestimating it costs damaged cargo. Replacing the rule by a fitted physics engine would raise the cost of a single evaluation from microseconds to about nine milliseconds (Mazur et al., 2025, p. 9), which is affordable offline but not inside an online placement loop that must answer per box (Mazur et al., 2022). The placement stage maximises geometric density and ignores the thermal function of the load. Postharvest engineering shows that the stacking pattern changes how fast and how uniformly a pallet cools: interlocked stacking, adopted for mechanical stability, blocks vent holes, and blocked cartons need two to three times longer to cool than upstream ones (Wu et al., 2019, p. 16). The binding variable is vent alignment rather than density as such: in the same study the least dense pallet design, holding 3,900 fruit against 5,120, is also the one that cools worst (Wu et al., 2019, pp. 5–6). Container trials report that stowage changes cooling rate and energy use, although the effect on mean cooling time is small (Defraeye et al., 2016), and for cut roses larger or overfilled boxes cool more slowly while cross-stacking improves airflow (Harkema et al., 2017, pp. 19–20). These studies concern sea-freight reefers, whereas air cargo relies on precooling on the ground and on hold temperature, with active containers as the controlled alternative, which for

this aircraft means the RAP e2 and RKN e1 containers (Baxter and Kourousis, 2015, p. 168); still, a stacking pattern that blocks vents could slow cooling and shift cost from airfreight to product quality, so the gain reported here is a logistics gain conditional on adequate precooling. Finally, all results concern a single historical manifest of one aircraft type; the cross-study replication of the policy ranking mitigates, but does not remove, this restriction.

8.2 Future work

Four extensions follow naturally. First, within-pallet weight: adding a heavy-below preference to the placement policies would carry the weight-awareness of Stage 1 into the vertical dimension of Stage 2. Second, explicit contoured geometry: modelling the fuselage-adjacent region of each main-deck position as a second, reduced-height bin (the dual-region decomposition prototyped in the precursor study) would close the remaining gap between nominal and prism capacities. Third, intra-day reoptimisation: because the aggregated Stage-1 model solves in under a second, the allocation can be re-run whenever the manifest changes after trucks depart, turning the DSM into a rolling planning tool. Fourth, and most valuable as evidence: repeating the full two-stage pipeline against the manual historical operation on the precursor’s sixteen-flight Boeing 747 dataset, with the ILP re-solved per flight, would provide the like-for-like multi-flight validation that neither precursor could offer alone.

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